Codec Operator Linear Algebra

The Six-Register Computational Substrate

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1 Introduction

Physics of reality is represented as linear algebra on six universal registers and related macros:

$$\mathbf{R} = \begin{bmatrix} \mathrm{Id} \\ \Phi \\ X \\ S \\ C \\ L \end{bmatrix} \in \mathbb{R}^6, \quad \mathbf{R}_{\mathrm{new}} = M_{\mathrm{codec}}(\mathrm{params})\,\mathbf{R}_{\mathrm{old}} \tag{1}$$

Each 6×6 matrix M_i represents an executable codec—a primitive physical operation acting on the substrate. Unless stated otherwise, all coefficients are real and dimensionless. These are not adjustable parameters but architectural constraints whose values follow from the finite-capacity structure. However, their operational thresholds and relative activation strengths constitute testable predictions.

2 The Nine Codec Matrices

2.1 Codec $M_1(\lambda_s)$: Scalar Emission (Higgs-like)

$$M_1(\lambda_s) = \begin{bmatrix} 1 & 0 & 0 & -\ell_1 & +\sigma_1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & (1-\lambda_s) & 0 & 0 \\ 0 & 0 & 0 & 0 & (1+\mu_1) & 0 \\ 0 & 0 & 0 & \eta_1 & 0 & 1 \end{bmatrix} \tag{2}$$

Entropy S decreases by $(1-\lambda_s)$, coherence C rises slightly, and a small load cost $\eta_1 S$ enters L. Represents scalar field relaxation and mass-generating mechanism.

2.2 Codec $M_2(f)$: Vector Emission (Photon/EM)

$$M_2(f) = \begin{bmatrix} 1 & 0 & 0 & -a_2 & +b_2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & d_2 & 0 \\ 0 & 0 & 0 & \varepsilon_2 f & 0 & 1 \end{bmatrix} \tag{3}$$

Entropy $\downarrow$, coherence $\uparrow$, small load increase $f\varepsilon_2$. Represents coherent photon emission. The ratio c_2/d_2 determines the entropy-coherence tradeoff during electromagnetic relaxation.

2.3 Codec $M_3(g, \kappa)$: Tensor Emission (Graviton)

$$M_3(g, \kappa) = \begin{bmatrix} 1 & 0 & 0 & -a_3 & +b_3 & 0 \\ 0 & (1+\alpha_3) & 0 & 0 & \beta_3 & 0 \\ 0 & 0 & (1+\gamma_3) & 0 & 0 & 0 \\ 0 & 0 & 0 & (1-\kappa g) & 0 & 0 \\ 0 & 0 & 0 & 0 & (1+\delta_3) & 0 \\ 0 & 0 & 0 & \varepsilon_3 g & 0 & 1 \end{bmatrix} \tag{4}$$

Infinitesimal entropy leakage to gravitational channel L with phase/spatial coupling. The extreme weakness of $g \sim 10^{-8}$ reflects the high computational cost (47 units) of tensor processing.

2.4 Codec $M_4(\theta_w, \Gamma_w)$: SU(2) Weak Interaction

$$M_4(\theta_w, \Gamma_w) = \begin{bmatrix} \cos\theta_w & \sin\theta_w & 0 & -\alpha_4 & 0 & 0 \\ -\sin\theta_w & \cos\theta_w & 0 & 0 & \beta_4 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & (1-\Gamma_w) & 0 & 0 \\ 0 & 0 & 0 & 0 & (1-\gamma_4) & 0 \\ 0 & 0 & 0 & \varepsilon_4 \Gamma_w & 0 & 1 \end{bmatrix} \tag{5}$$

Implements parity-violating Identity–Phase rotation and entropy dump Γ_w. The mixing angle θ_w determines the strength of flavor-changing processes.

2.5 Codec $M_5(\Lambda_s, \chi)$: SU(3) Strong Interaction

$$M_5(\Lambda_s, \chi) = \begin{bmatrix} 1 & 0 & 0 & -\alpha_5 & +\beta_5 & 0 \\ 0 & (1+\rho_5) & 0 & 0 & \sigma_5 & 0 \\ 0 & 0 & (1+\tau_5) & 0 & \upsilon_5 & 0 \\ 0 & 0 & 0 & (1-\Lambda_s) & 0 & 0 \\ 0 & 0 & 0 & 0 & (1+\chi) & 0 \\ 0 & 0 & 0 & \varepsilon_5 \Lambda_s & 0 & 1 \end{bmatrix} \tag{6}$$

Strong local re-coherence and spatial–phase mixing; large entropy consumption when S exceeds the strong threshold. Activation requires high local density (short-range confinement).

2.6 Codec $M_6(S)$: Identity Rewrite (Pauli/Tensor Swap)

$$M_6(S) = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_6 & 0 \\ 0 & 0 & 0 & S & 0 & 1 \end{bmatrix}, \quad \text{apply if } S > S_{\text{thr}} \tag{7}$$

Flips identity, consumes entropy, reduces coherence ($C \to \rho_6 C$), dumps load $+S$ to L. This is the Pauli exclusion mechanism: when two registers collide in state space ($S > S_{\text{thr}}$), one must rewrite its identity.

2.7 Codec $M_7(\omega, \phi_0)$: Phase Rotation

$$M_7(\omega, \phi_0) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cos(\omega\Delta t + \phi_0) & -\sin(\omega\Delta t + \phi_0) & 0 & \alpha_7 & 0 \\ 0 & \sin(\omega\Delta t + \phi_0) & \cos(\omega\Delta t + \phi_0) & 0 & \beta_7 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & (1+\gamma_7) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{8}$$

Pure phase–space rotation with small coherence coupling. Implements unitary time evolution in the coherent regime.

2.8 Codec $M_8(v, \Delta t)$: Spatial Translation

$$M_8(v, \Delta t) = \begin{bmatrix} 1 & 0 & 0 & -\alpha_8 v\Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \beta_8 v\Delta t & (1+\gamma_8) & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \varepsilon_8 v\Delta t & 0 & 0 & 1 \end{bmatrix} \tag{9}$$

Advects spatial register, producing small entropy increase and load work. Kinetic energy emerges from accumulated load L during spatial translation.

2.9 Codec $M_9(\Delta S)$: Entropy Increase (Second-Law Drive)

$$M_9(\Delta S) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & (1+\Delta S) & 0 & 0 \\ 0 & 0 & 0 & 0 & (1-\kappa_9\Delta S) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

Implements dissipative heating: S grows, C decreases by $\kappa_9\Delta S$. Represents thermalization and decoherence.

3 Execution Rule

At each time step, apply codecs in the causal sequence of active channels. Conditional codecs (M_6, possibly M_3, M_5) act only when their trigger inequalities are satisfied. Non-commuting codecs encode order-sensitive phenomena such as emission-before-flip versus flip-before-emission behavior.

Symbolic summary:

$$\mathbf{R}_{\text{new}} = \prod_{i \in \mathcal{A}} M_i(\text{params}_i)\, \mathbf{R}_{\text{old}}, \quad [M_i, M_j] \neq 0 \Rightarrow \text{path-dependent evolution} \tag{11}$$

4 Testable Operational Thresholds

The codec algebra makes no claims about adjustable constants. Instead, it predicts threshold ratios and activation orderings that are experimentally testable:

Table 1: Codec activation thresholds and testable relationships. These are architectural predictions, not fitted parameters.

Codec	Activation Condition	Testable Prediction
M_1 scalar	Continuous (always active)	Higgs coupling ratios
M_2 vector	$L > L_{\text{EM}}$	EM emission threshold
M_3 tensor	$\lvert\partial^2\Phi/\partial r^2\rvert^{-1} > K_{\text{cap}}$	ISCO spike at 10.7×
M_4 weak	Flavor mismatch	Weak decay rates
M_5 strong	$S > S_{\text{strong}}$, $r < r_{\text{confine}}$	Confinement scale
M_6 rewrite	$S > S_{\text{thr}}$ (Pauli collision)	Exclusion spectroscopy
M_7 phase	Coherent regime ($C > 0.5$)	Unitary evolution
M_8 spatial	$v > 0$ (motion)	Kinetic energy accumulation
M_9 entropy	Stochastic/thermal drive	Decoherence timescales

4.1 Experimental Tests

1. **Threshold ordering:** Predict $S_{\text{thr}} < S_{\text{strong}}$ (Pauli activates before confinement)
2. **Capacity saturation:** M_3 activation scales as $L \propto |\partial^2\Phi/\partial r^2|^{-1}$ (ISCO test)
3. **Entropy-coherence tradeoff:** Measure $c_2 \approx 0.90$, $d_2 \approx 1.05$ via cavity emission spectroscopy
4. **Rewrite coherence loss:** $\rho_6 \approx 0.70$ testable through Pauli-forbidden transition suppression
5. **Non-commutativity:** $[M_2, M_6] \neq 0$ predicts order-dependent emission spectra

These are architectural constraints, not free parameters. Their values follow from the 1:10:47 cost hierarchy and the 10^6 packet-per-tick capacity. Deviation from these predictions would falsify the codec algebra, not merely require parameter adjustment.

5 Interpretation

Every physical interaction corresponds to a matrix product on the six registers. The Standard Model's gauge structure (U(1) × SU(2) × SU(3)) emerges as the algebra of codec matrices that preserve certain register invariants under composition.

Crucially:

- **Fermions** saturate registers (one per channel) → Pauli exclusion via M_6
- **Bosons** accumulate coherently (broadcast mode) → no M_6 trigger
- **Weak interactions** violate parity via M_4's asymmetric mixing
- **Strong interactions** confine via M_5's threshold activation
- **Gravity is weak** because M_3 has computational cost 47 (tensor processing)

The universe does not "choose" gauge groups—it executes the algebra of matrices that preserve register integrity under finite capacity.